Cheat sheets

Unit 1: Multimedia System

Formulas and facts

- Multimedia = Computer-controlled + Integrated + Digital + Interactive (CIDI)
- Discrete: text, image, graphics. Continuous: audio, video, animation.

Memory hooks

- A true multimedia system is Computer-controlled, Integrated, Digital and Interactive: remember it as the word CIDI.
- Five jobs to remember: generate, represent, store, transmit, search and retrieve; compression and processing sit across all of them.
- Three domains and one bar: Application, System, Device, with Synchronization running through all three.
- Six categories: Perception, Representation, Presentation, Storage, Transmission, Exchange. Memory sentence: Please Read Pages, Study Thoroughly Every day.
- Discrete = space only. Continuous = space + time. Three properties: combination, independence, computer-supported integration.
- Asynchronous = whenever. Synchronous = not later than a limit. Isochronous = not earlier and not later: a steady rhythm.

Key terms

Medium	A means of distributing and presenting information (text, graphics, speech, music, video).
Multimedia	Computer-controlled integration of several media, represented digitally and usually interactive.
Discrete medium	Time-independent: text, still images, graphics.
Continuous medium	Time-dependent: audio, video, animation.
Hypertext	Text organised as linked nodes that the reader can follow in any order.
Non-linear	The user, not the author, chooses the path through the content.
Multimedia system	A system that processes multimedia data and applications.
Hardware	Capture, storage, processing and presentation devices.
Software	Operating system, drivers and application programs.
Network	Carries media between systems, usually with real-time needs.
Compression	Reducing the amount of data needed to represent a medium.

Application domain	Tools and applications that users work with; provides the user interface.
System domain	Database systems, operating system and communication services.
Device domain	Physical devices: capture, storage, input and output hardware.
Synchronization	Keeps media in the right time relation across all domains.
Perception medium	How humans sense information: seeing and hearing.

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Unit 2: Sound and Audio

Formulas and facts

- Nyquist: fs ≥ 2·fmax
- Size (bytes) = rate × bits/8 × channels × seconds
- CD-DA = 44,100 × 16 × 2 = 1,411,200 bit/s
- Levels = 2^bits

Memory hooks

- Amplitude decides loudness, frequency decides pitch. Humans hear about 20 Hz to 20 kHz. Frequency f = 1 / T.
- Sample at least twice the highest frequency (Nyquist). Size = rate × (bits/8) × channels × seconds.
- MIDI files are tiny and easy to edit, but they need a synthesizer to be heard. Note On = status byte + key number + velocity.
- Who = verification and identification. What = recognition and understanding. How = emotion and state of the speaker.
- Two steps of synthesis: text is transcribed into a sound script, then the sounds are synthesised into speech.
- PCM = accurate but big. Source encoding = smaller. Recognise-then-synthesise = smallest, but the original voice is lost.

Key terms

Sound	Mechanical wave energy that travels through a medium.
Audio	Electrical signal (analog or digital) that represents sound.
Amplitude	Height of the wave: loudness.
Frequency	Cycles per second (Hz): pitch.
Period T	Time of one full cycle; T = 1/f.
Infrasound / ultrasound	Below 20 Hz / above 20 kHz: not heard by humans.
Sampling rate	Number of samples per second, in Hz.
Bit depth	Bits per sample; 16 bits gives 65,536 levels.
Quantization	Rounding a sample to the nearest allowed level.
PCM	Pulse Code Modulation: samples written as binary codes.
Nyquist theorem	fs must be at least 2 × fmax to avoid aliasing.
Channel	Mono = 1 channel, stereo = 2.
MIDI	Musical Instrument Digital Interface: hardware and message standard (early 1980s).
Note On / Note Off	Messages that start and stop a note.

Velocity	How hard a key is struck (0 to 127).
Synthesizer	Device or software that makes sound from MIDI messages.

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Unit 3: Image and Graphics

Formulas and facts

- Bitmap bytes = W × H × bits / 8
- 640×480×8-bit = 307,200 bytes
- Dithering trades resolution for apparent colour depth

Memory hooks

- Bitmap size = width × height × bits per pixel ÷ 8. Bitmaps store pixels; vector graphics store shapes.
- Photos use JPEG. Sharp-edged graphics use PNG or GIF. TIFF is for high-quality storage.
- Recognition pipeline: capture, pre-process, find features, label, interpret.
- Transmission time = size in bits ÷ link rate. A 640 × 480 8-bit image is 307,200 bytes: about 38.4 s at 64 kbit/s.
- Flow: host sends display commands, display controller draws into the refresh buffer, video controller reads the buffer and drives the monitor.

Key terms

Pixel	Picture element: the smallest unit of an image.
Resolution	Number of pixels, for example 640 × 480.
Bit depth	Bits per pixel: 1 = black/white, 8 = 256 values, 24 = true colour.
Bitmap	Image stored pixel by pixel.
Vector graphic	Picture stored as shapes and formulas.
RGB / CMY	Additive screen colours / subtractive print colours.
BMP	Simple, usually uncompressed bitmap.
GIF	Up to 256 colours, lossless, supports animation.
JPEG	Lossy compression suited to photographs.
PNG	Lossless compression with true colour and transparency.
TIFF	Flexible, high-quality format used in publishing and scanning.
Lossless vs lossy	Exact copy after decoding / some information lost.
Image analysis	Extracting information from images.
Image recognition	Deciding what the image or an object in it is.
Pre-processing	Cleaning the image: noise removal, contrast.
Segmentation	Splitting the image into regions or objects.

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Unit 4: Video and Animation

Formulas and facts

- Y = 0.30R + 0.59G + 0.11B; U ∝ (B−Y); V ∝ (R−Y)
- 25 fps → 40 ms per frame
- Raw 640×480×24-bit×25 fps = 184.32 Mbit/s

Memory hooks

- 25 frames per second means 40 ms per frame. PAL: 625 lines, 25 fps. NTSC: 525 lines, about 30 fps.
- Data rate = width × height × bytes per pixel × frames per second. CIF is 352 × 288.
- Y = 0.30R + 0.59G + 0.11B. The eye is more sensitive to brightness than to colour, so colour can use less bandwidth.
- Key frames define the motion, in-betweens (tweening) fill the gaps. Film uses about 24 frames per second.
- Describe the motion, not the pictures: this saves effort and data.
- Double buffering: draw the next frame off-screen, then swap. Symbolic transmission is smaller than pixel transmission.

Key terms

Frame	One complete still image of a video.
Frame rate	Frames shown per second.
Interlacing	Sending odd lines then even lines (two fields) to reduce flicker.
Luminance (Y)	Brightness part of the signal.
Chrominance	Colour information.
Aspect ratio	Width to height, for example 4:3 or 16:9.
Digitizer	Converts analog video into digital frames.
Resolution	Pixels per frame.
Chroma subsampling	Storing colour at lower resolution than brightness.
CIF	Common Intermediate Format: 352 × 288 pixels.
RGB	Three separate colour signals.
YUV	Luminance plus two colour-difference signals (PAL).
YIQ	Luminance plus in-phase and quadrature signals (NTSC).
Composite	All components combined in one signal.
Digitization	Sampling and quantizing the video signal.
Key frame	A frame that defines an important pose or position.

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Unit 5: Data Compression

Formulas and facts

- Ratio = original : compressed
- Avg length = $\Sigma\, p_i \cdot l_i$
- RLE: aaabccdddd → a3b1c2d4
- JPEG: 8×8 → DCT → quantize (lossy) → zigzag → RLE/Huffman

Memory hooks

- Compression ratio = original size ÷ compressed size. Dialogue applications need a delay below about 150 ms.
- Entropy = ignores meaning. Source = uses meaning. Hybrid = both.
- Huffman: repeatedly merge the two smallest frequencies; left edge 0, right edge 1; no code is a prefix of another.
- Send the difference, not the value: small numbers need fewer bits.
- Pipeline: colour transform, 8 × 8 blocks, DCT, quantization, zig-zag, run-length / differential, Huffman.
- I frames stand alone, P frames are predicted from the past, B frames from the past and the future.

Key terms

Redundancy	Repeated or predictable information that can be removed.
Lossless	Original is recovered exactly.
Lossy	Some information is permanently lost.
Compression ratio	Original size : compressed size.
Codec	Coder and decoder pair.
Symmetric coding	Encoding and decoding take similar effort.
Entropy coding	Lossless; treats data as a bit sequence: RLE, Huffman, arithmetic.
Source coding	Uses properties of the medium: prediction, transform, subband.
Hybrid coding	Combines techniques: JPEG, MPEG, H.261, DVI.
Run-length coding	Replaces runs with value and count.
DCT	Discrete Cosine Transform used in JPEG and MPEG.
Run	Consecutive repeats of the same value.
Prefix code	No code word is the start of another, so decoding is unambiguous.
Variable-length code	Code words of different lengths.
Priority queue	List kept sorted by frequency to find the two smallest.

Frequency	How often a symbol occurs.

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Unit 6: Optical Storage Media

Formulas and facts

- Block = 2,352 bytes, 75 blocks/s
- 74 min × 60 × 75 = 333,000 blocks
- Mode 1 user data 2,048 B (≈650 MB); Mode 2 2,336 B
- Lasers: CD 780 nm, DVD 650 nm, Blu-ray 405 nm

Memory hooks

- Three layers: polycarbonate substrate, reflective layer, protective layer. A change between pit and land represents a 1.
- Shorter wavelength + smaller pits + tighter tracks = more capacity: CD 780 nm, DVD 650/635 nm, Blu-ray 405 nm.
- Red Book. 44,100 × 16 × 2 = 1,411,200 bit/s. 74 min = 333,000 blocks of 2,352 bytes.
- Yellow Book. Mode 1 = 2,048 user bytes + EDC/ECC. Mode 2 = 2,336 user bytes.
- Photo CD resolutions: Base/16, Base/4, Base, 4 Base, 16 Base. CD-I = Green Book.
- Orange Book covers both: WO = write once, MO = magneto-optical rewritable.

Key terms

Pit / land	Recessed / flat areas of the track.
Spiral track	One track winding from inside to outside.
Laser	Light source used to read (and write) the disc.
Substrate	Clear polycarbonate plastic carrying the pits.
CLV	Constant Linear Velocity: the disc speed varies so data passes at a constant rate.
Optical storage	Storage read with light.
LaserDisc	Analog video disc using laser reading.
DVD	Digital Versatile Disc: 4.7 GB single layer.
Blu-ray	Uses a blue-violet 405 nm laser; 25 GB per layer.
Track pitch	Distance between tracks: 1.6 µm CD, 0.74 µm DVD.
Dual layer	Two data layers read at different focus.
Red Book	Specification of the audio CD.
EFM	Eight-to-Fourteen Modulation: line code for the disc.
CIRC	Cross-Interleaved Reed-Solomon Code for error correction.
Subcode	Extra channel with track and time information.
Block / sector	2,352 bytes; 75 per second.

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Unit 7: Multimedia Operating System

Formulas and facts

- $U = \Sigma C_i/P_i$
- EDF: schedulable iff $U \leq 1$ ($D = P$)
- RM bound: $n(2^{(1/n)} - 1)$; sufficient, not necessary
- Example (2,5)+(4,7): $U = 0.971$ → EDF yes, RM no

Memory hooks

- Hard real time: a missed deadline is a failure. Soft real time: a miss reduces quality.
- Reserve first, then guarantee. LBAP model: $B + R \cdot t$, the data that may arrive by time t.
- Utilisation $U = \Sigma C/P$. EDF works if $U \leq 1$. RM is guaranteed if $U \leq n(2^{(1/n)} - 1)$.
- Goal: guarantee timing. Layers: device, driver, kernel scheduler, file system, application interface.

Key terms

Deadline	Latest time by which a task must finish.
Hard real time	Missing a deadline is unacceptable.
Soft real time	Occasional misses degrade quality only.
Periodic task	Task that repeats with a fixed period.
Jitter	Variation in timing.
Resource	Anything a stream needs: CPU, memory, bandwidth, disk.
Admission control	Accepting or rejecting a new request.
Reservation	Setting resources aside in advance.
LBAP	Linear Bounded Arrival Process model.
Utilisation	Fraction of a resource in use.
Period P	Time between releases of a periodic task.
Computation time C	CPU time needed per period.
Utilisation U	Sum of C/P over all tasks.
EDF	Earliest Deadline First: dynamic priority, optimal.
RM	Rate Monotonic: static priority by period.
Preemption	Interrupting a running task for a more urgent one.

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Unit 8: Multimedia Communication

Formulas and facts

- Leaky bucket: constant output rate, overflow dropped
- RTP: sequence no. + timestamp; RTCP: feedback
- QoS: negotiate → admit → guarantee

Memory hooks

- Needs: throughput, low delay, low jitter, tolerable loss, multicast.
- QoS parameters: throughput, delay, jitter, loss. Shaping uses a leaky or token bucket.
- UDP + RTP for media, RTCP for feedback, TCP is a poor fit for real time.
- Collaborative computing + session management + architecture + mail and messaging.

Key terms

Bandwidth	Data capacity of a link.
Delay (latency)	Time for data to travel from sender to receiver.
Jitter	Variation of delay.
Loss	Fraction of data that does not arrive.
Multicast	One sender, many receivers.
QoS	Quality of Service guarantees.
Guaranteed / best effort	Firm promise / no promise.
Negotiation	Agreeing a QoS between user and system.
Token bucket	Allows bursts up to b and rate r.
Traffic shaping	Smoothing the flow to match the agreed profile.
RTP	Real-time Transport Protocol.
RTCP	RTP Control Protocol: quality reports.
UDP	Fast, connectionless transport without retransmission.
Sequence number	Detects loss and reordering.
Timestamp	Tells the receiver when to play a packet.
Collaborative computing	Several users working together using computers.

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Unit 9: Documents, Hypertext and MHEG

Formulas and facts

- Hypermedia = hypertext + multimedia
- Dexter layers: Run-time / Storage / Within-component
- SGML: structure via DTD; ODA: logical + layout; MHEG: final-form objects

Memory hooks

- Content, structure, layout: keep them separate for flexibility.
- Hypertext = nodes + links + anchors. The Dexter model describes the layers.
- Remember the line: Memex (1945), hypertext term (1965), Web (1989 to 1991).
- SGML = markup with a DTD. ODA = logical + layout structure. MHEG = multimedia objects for exchange.

Key terms

Document	Structured collection of information, shown to a reader.
Logical structure	Organisation into chapters, sections, paragraphs.
Layout structure	How content is placed on the page or screen.
Content	The actual text, images, audio.
Document architecture	Model describing how a document is built.
Node	A unit of information.
Link	A connection between nodes or anchors.
Anchor	The active spot inside a node where a link starts or ends.
Hypermedia	Hypertext with audio, video and images.
Dexter model	Reference model for hypertext systems.
Memex	Vannevar Bush's 1945 idea of a linked personal information machine.
Hypertext	Term coined by Ted Nelson in 1965.
World Wide Web	Proposed by Tim Berners-Lee in 1989; first web site in 1991.
Browser	Program that displays and follows links.
Authoring system	Tool used to create hypermedia.
SGML	Standard Generalized Markup Language, ISO 8879.

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Unit 10: Synchronization

Formulas and facts

- Lip sync: skew within ±80 ms
- Drift = clock error × time (0.1% × 600 s = 0.6 s)
- Fix: clock sync, buffering, adaptive playout

Memory hooks

- Intra = smooth within a stream. Inter = aligned between streams. Live = real time source, synthetic = pre-planned.
- Four layers: media, stream, object, specification.
- Lip sync tolerance: audio may be 80 ms behind to 80 ms ahead of the video.
- Four common categories: interval-based, axes (timeline) based, control-flow based, event-based.
- Delay variation is removed by buffers; clock drift by clock sync.

Key terms

Intra-object sync	Correct timing inside one medium.
Inter-object sync	Correct timing between different media.
Live synchronization	Sync of live, captured media.
Synthetic synchronization	Planned timing of stored media.
Hard / soft sync	Strict / tolerant timing relation.
Skew	Time difference between streams.
Media layer	Single continuous stream of LDUs.
Stream layer	Groups of streams and their sync.
Object layer	Complete multimedia objects, including discrete media.
Specification layer	Author's description of the presentation.
Reference model	Structured view that guides system design.
Lip synchronization	Matching speech to lip movement.
Tolerance	The error people still accept.
Jitter	Short-term delay variation.
Presentation requirement	Timing limit that quality depends on.
Specification	Formal description of timing relations.

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Unit 11: Abstraction of Programming

Formulas and facts

- Abstraction levels: device/system → library → toolkit → language → application
- Toolkit = reusable object components

Memory hooks

- Low to high: device, library, system software, toolkit, higher language, object-oriented.
- Simple objects (BMO), simple multimedia objects (SMO) and compound multimedia objects (CMO).
- Two models: data as time capsules, data as streams.
- Toolkit = ready-made parts. Language = media as data types (for example OCCAM-2 with parallel processes).

Key terms

Abstraction level	Degree of detail used to represent and manipulate data.
Library	Functions supplied with a device.
Toolkit	Uniform interface that hides process structure.
Programming language	Language with abstractions for media.
Object-oriented approach	Class hierarchy for multimedia.
Class	Blueprint of an object.
Object	An instance of a class.
Inheritance	A class gets features from a parent class.
BMO	Basic Media Object.
SMO / CMO	Simple / Compound Multimedia Object.
Stream	Flow of data units from a source to a sink.
Time capsule	Data stored with its timing information.
Nemo system	Example of multimedia system software.
Scheduler	Allocates CPU time.
Driver	Controls hardware.
API	Application Programming Interface.

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Unit 12: Multimedia Application

Formulas and facts

- Development cycle: plan → design (storyboard) → produce → test → deliver
- Systems: input + processing + storage + output components
- Applications: VoD, video conferencing, education, information systems, digital libraries, entertainment

Memory hooks

- VoD needs a server, a network and a set-top box. Conferencing needs low delay.
- Six classes: games, repositories, iTV/VoD, conferencing/mail, shared workspaces, business workflow.
- Plan → Design → Produce → Test → Deliver, and always check copyright.
- Three component groups: input, output, storage.

Key terms

Video on demand	Watch a chosen video when you want.
Video conferencing	Live audio-video link between sites.
Set-top box	Connects the TV to the digital network.
Computer-based training	Learning delivered by computer.
Information system	System for storing and retrieving information.
Game systems	Leaders in multimedia technology.
Repository	Large indexed database, mostly play-back.
Interactive TV	TV with interactive services.
Shared workspace	Remote users see the same application.
Workflow	Business process using multimedia data.
Storyboard	Sequence of sketches showing the screens and flow.
Prototype	Early working version.
Rights and permissions	Legal permission to use others' content.
Testing	Finding and fixing problems.
Delivery	Publishing the product to users.
Input systems	Devices that capture media: camera, microphone, scanner.

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